Revision | **Covers** 4-1 → 4-5 | **File** 7 / 11

FS

Master formula sheet

ورقة القوانين الشاملة — احفظها كاملة

Everything in Chapter 4 on one page. If you can reproduce this from memory in four minutes, you are ready.

كيف تستخدم ورقة القوانين

اقرأها حسب القسم (الألوان نفسها في الملخص)، والعمود الأخير يخبرك **متى** تستخدم كل قانون، وتحت كل اسم إنجليزي اسمه بالعربي. التدريب الحقيقي: اكتبها من ذاكرتك على ورقة فارغة، ثم قارن وصحّح بالأحمر. إذا استطعت كتابتها كاملة في أربع دقائق فأنت جاهز.

NAME	FORMULA	WHEN
§4-1 · Sample spaces and probability		
Classical probability الاحتمال الكلاسيكي	$P(E) = n(E) / n(S)$	Equally like outcomes; you can cou the sample space.
Empirical probability الاحتمال التجريبي	$P(E) = f / n$	A frequency distribution or survey data is give
Probability range مدى الاحتمال	$0 \le P(E) \le 1$	Always. Use to sanity-check every answer.
Total probability مجموع الاحتمالات	$\Sigma P(\text{all outcomes in } S) = 1$	Check on ar frequency c tree table.

NAME	FORMULA	WHEN
Complement rule قاعدة المتمّمة	$P(\bar{E}) = 1 - P(E)$	"not", "neither", "none", "at least one".
§4-2 · Addition rules — "OR"		
Mutually exclusive test اختبار التنافي	$A \cap B = \varnothing$	Can the two events happen together? N → mutually exclusive.
Addition rule 1 قاعدة الجمع الأولى	$P(A \cup B) = P(A) + P(B)$	A and B **are** mutually exclusive.
Addition rule 2 قاعدة الجمع الثانية	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	A and B are **not** mutuall exclusive. Safe in both cases.
Three mutually exclusive ثلاثة أحداث متنافية	$P(A \cup B \cup C) = P(A) + P(B) + P(C)$	Three disjoi events.
Three general ثلاثة أحداث غير متنافية	$P(A \cup B \cup C) = \Sigma P(\text{single}) - \Sigma P(\text{pairs}) + P(A \cap B \cap C)$	Three overlapping events.
§4-3 · Multiplication rules — "AND" and "GIVEN"		

NAME	FORMULA	WHEN
Multiplication rule 1 قاعدة الضرب الأولى	$P(A \cap B) = P(A) \cdot P(B)$	A and B independer (with replacemer separate devices).
Multiplication rule 2 قاعدة الضرب الثانية	$P(A \cap B) = P(A\|B) \cdot P(B) = P(B\|A) \cdot P(A)$	A and B dependent (without replacemer "of those who...").
Conditional probability الاحتمال الشرطي	$P(B \mid A) = P(A \cap B) / P(A)$	"given that" "if", "of thos who", a stated accomplish fact.
Independence test اختبار الاستقلال	$P(B\|A) = P(B)$ or $P(A \cap B) = P(A) \cdot P(B)$	Asked directly, especially from a two-way table.
At least one واحد على الأقل	$P(\text{at least one}) = 1 - P(\text{none})$	Any questic containing " least".
§4-4 & §4-5 · Counting rules		
Fundamental counting rule قاعدة العدّ الأساسية	$k_1 \times k_2 \times \cdots \times k_n$	A sequence of independer choices; repetition allowed or not.

NAME	FORMULA	WHEN
Factorial المضروب	$n! = n(n-1)(n-2)\cdots(2)(1), \quad 0! = 1$	Arranging a n objects.
Permutation rule 1 التباديل	${}_nP_r = n! / (n - r)!$	Choose r from n, **ord matters**.
Permutation rule 2 التباديل مع عناصر متكررة	$n! / (r_1!\ r_2!\ \cdots\ r_p!)$	Arrange n objects, some identical (letters, identical items).
Combination rule التوافيق	${}_nC_r = n! / [(n - r)!\ r!]$	Choose r from n, **ord does not matter**.
Link between them العلاقة بينهما	${}_nC_r = {}_nP_r / r!$	Sanity chec C is always P.
Probability by counting الاحتمال بالعدّ	$P(E) = n(E) / n(S)$, both counted with the rules above	**The signature § 5 question.**

Casio calculator keystrokes

TO COMPUTE	PRESS	RESULT
$(5 + 3)!$	`( 5 + 3 )  SHIFT  x⁻¹ (x!)  =`	40,320
${}_7P_4$	`7  SHIFT  × (nPr)  4  =`	840
${}_{10}C_4$	`10  SHIFT  ÷ (nCr)  4  =`	210

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet ← this file

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.